

# IMO Training – Number Theory

## IMO Shortlist Problems

Longtin Chan  
longtinclt@gmail.com

16 January, 2020

### Notes

Disum First.

$10 \sim 1$  14:30.

$12 \sim 2$

$13 \sim 1$

$14 \sim 2$

$15 \sim 1$

$19 \sim 2$

Solve Equation  
Function.

⋮

$11 \sim 1$

$16 \sim 1$

$17 \sim 1$

$18 \sim 1$

Unconventional

- construction.

⋮

1. (IMO 2010 N1)

Find the least  $n \in \mathbb{N}$  for which  $\exists$  a set  $\{s_1, s_2, \dots, s_n\}$  consisting of  $n$  distinct positive integers, such that

$$\left(1 - \frac{1}{s_1}\right) \left(1 - \frac{1}{s_2}\right) \cdots \left(1 - \frac{1}{s_n}\right) = \frac{51}{2010}.$$

$$s_i \geq 2$$

$$\frac{1}{s_i} \leq \frac{1}{2}$$

$$-\frac{1}{s_i} \geq -\frac{1}{2}$$

Notice that

$$\begin{aligned} & \left(1 - \frac{1}{s_1}\right) \left(1 - \frac{1}{s_2}\right) \cdots \left(1 - \frac{1}{s_n}\right) \\ & \geq \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \cdots \left(1 - \frac{1}{n+1}\right) \\ & = \frac{1}{n+1}. \end{aligned}$$

$$\therefore \frac{51}{2010} \geq \frac{1}{n+1}.$$

$$n+1 \geq \frac{670}{17}$$

$$n+1 \geq 39 \frac{7}{17}.$$

$$n \geq 38 \frac{7}{17}$$

$$\therefore n \geq 39.$$

$$\begin{aligned} & \therefore \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \cdots \left(1 - \frac{1}{33}\right) \\ & \quad \left(1 - \frac{1}{35}\right) \cdots \left(1 - \frac{1}{40}\right) \left(1 - \frac{1}{67}\right) \end{aligned}$$

$$= \frac{1}{\cancel{33}^{17} \cdot \cancel{40}^{17} \cdot 67} \cdot \cancel{66}^2$$

$$= \frac{17}{670}$$

$$= \frac{51}{2010}$$

$\therefore$  The required minimum value of  $n$  is 39.

## 2. (IMO 2011 N1)

For any  $d \in \mathbb{N}$ , let  $f(d)$  be the smallest possible integer that has exactly  $d$  positive divisors (e.g.  $f(1) = 1, f(5) = 16, f(6) = 12$ ). Prove that for every  $k \in \mathbb{Z}_{\geq 0}$ ,  $f(2^k) \mid f(2^{k+1})$ .

3. (IMO 2012 N2)

Find all triples  $(x, y, z)$  of positive integers such that  $x \leq y \leq z$  and  $\text{valuation} \leq 3$ .

$$x^3(y^3 + z^3) = 2012(xyz + 2). = 2012xyz + 4024.$$

$2012 = 2^3 \cdot 503$

$1, 2, 4, 8$

$$\cdot \text{Mod } x \Rightarrow 4024 \equiv 0 \pmod{x}, \text{ i.e. } x \mid 2^3 \cdot 503.$$

$$\cdot \text{Check whether } 503 \mid x. \quad (\text{This is actually wrong!})$$

$$\cdot \text{Check 2-adic valuation} \Rightarrow x = 1, 2.$$

$$\cdot y^3 + z^3 = (y+z)(y^2 - yz + z^2) \quad y, z \not\equiv 0 \pmod{503}.$$

503 divides this?

$$y^3 + z^3 \equiv 0 \pmod{503}.$$

$$y^3 \equiv -z^3 \pmod{503}$$

$$y^{501} \equiv -z^{501} \pmod{503}$$

$$\therefore y^{502} \equiv -z^{502} \pmod{503}. \quad (\text{F.T.T.})$$

$$\therefore y \equiv -z \pmod{503}.$$

$$y + z \equiv 0 \pmod{503}.$$

$$\begin{aligned} \text{if } 503 \mid y^2 - yz + z^2, \\ \text{then } 503 \mid 4(y^2 - yz + z^2) \\ 503 \mid (2y - z)^2 + 3z^2. \end{aligned}$$

$$\therefore ((2y - z)z^{-1})^2 \equiv -3 \pmod{503}.$$

$$\begin{aligned} \text{However, } \left(\frac{-3}{503}\right) &= \left(\frac{-1}{503}\right) \left(\frac{3}{503}\right) \\ &= -\left(\frac{3}{503}\right) \\ &= -\left(\frac{503}{3}\right) (-1)^{\frac{503-1}{2} \cdot \frac{3-1}{2}} \\ &= \left(\frac{503}{3}\right) \\ &= \left(\frac{2}{3}\right) \\ &= -1. \end{aligned}$$

$$\therefore 503 \nmid y^2 - yz + z^2, \text{ and } 503 \mid y + z.$$

When  $x=1$ , the eq<sup>n</sup> becomes

$$y^3 + z^3 = 2012(yz + 2).$$

$$(y+z)(y^2 - yz + z^2) = 2012(yz + 2)$$

Let  $y+z = 503k$ ,  $k \in \mathbb{N}$ , then

$$k(y^2 - yz + z^2) = 4(yz + 2).$$

By AM-GM,  $y^2 - yz + z^2 \geq yz$ .

$$\therefore 4(yz + 2) \geq kyz.$$

$$k = 1, 2, 4.$$

 $\therefore$  RHS even, we have  $y^3 + z^3$  even,i.e.  $y+z$  even.

$$\text{When } k=2, \quad y^2 - yz + z^2 = 2yz + 4$$

$$y^2 - 3yz + z^2 = 4.$$

$$(y+z)^2 - 5yz = 4.$$

$$5yz = 1006^2 - 4. \quad \#$$

$$\text{When } k=4, \quad y^2 - yz + z^2 = yz + 2$$

$$(y-z)^2 = 2. \quad \#$$

When  $x=2$ , the eq<sup>n</sup> becomes

$$f(y^2+z^2) = 2012(2yz+2).$$

$$y^2+z^2 = 503(yz+1).$$

$$(y+z)(y^2-yz+z^2) = 503(yz+1).$$

$$k(y^2-yz+z^2) = yz+1.$$

Trivially,  $y^2-yz+z^2 \geq yz$ .

$$\therefore yz+1 \geq kyz, \text{ so } k=1$$

$$\therefore y^2-yz+z^2 = yz+1, \quad y+z=503$$

$$(y-z)^2 = 1$$

$$y-z = -1$$

$$\therefore y = 251, \quad z = 252$$

$$\therefore (x, y, z) = (2, 251, 252).$$

Checking:  $x^3(y^2+z^2) = f(503)(252^2 - 252 \cdot 251 + 251^2)$

$$= 4024(252+251^2)$$

$$2012(xy+2) = 2012(2 \cdot 251 \cdot 252 + 2)$$

$$= 4024(251 \cdot 252 + 1)$$



4. (IMO 2013 N1)

Find all functions  $f : \mathbb{N} \rightarrow \mathbb{N}$  such that *for all  $m, n \in \mathbb{N}$ ,*

$$m^2 + f(n) \mid mf(m) + n.$$

*if  $m = f(n)$ , then  $(f(n))^2 + f(n) \mid f(n)f(f(n)) + n$   
 $f(n)(f(n) + 1) \mid f(n)f(f(n)) + n$ .*

*In particular:  $f(n) \mid f(n)f(f(n)) + n$ .*

*$f(n) \mid f(n)f(f(n)) + n - f(n)(f(f(n)))$ .*

*$f(n) \mid n$ , ~~or~~ i.e.  $f(n) \leq n$  for all  $n$ .*

*if  $n = m$ , then  $m^2 + f(m) \mid mf(m) + m$ .*

$$m^2 + f(m) \leq mf(m) + m.$$

$$m^2 - m - mf(m) + f(m) \leq 0.$$

$$(m - 1)(m - f(m)) \leq 0.$$

$$1 \leq m \leq f(m).$$

*Combining,  $n = f(n)$  for all  $n$ .*

5. (IMO 2014 N2)

Find all pairs  $(x, y)$  of positive integers such that

$$\sqrt[3]{7x^2 - 13xy + 7y^2} = |x - y| + 1.$$

WLOG,  $x \geq y$ . Then  $7x^2 - 13xy + 7y^2 = (x - y + 1)^3$ .

Let  $x - y = d$ ,  $d \geq 0$ . The equation becomes

$$7(y+d)^2 - 13y(y+d) + 7y^2 = (d+1)^3.$$

$$y^2 + 14dy + 7d^2 - 13dy = (d+1)^3$$

$$y^2 + dy + 7d^2 - (d+1)^3 = 0.$$

$$\Delta = \square.$$

$$d^2 - 4(7d^2 - (d+1)^3) = \square.$$

$$d^2 - 4(7d^2 - (d^3 + 3d^2 + 3d + 1)) = \square$$

$$4d^3 - 15d^2 + 12d + 4 = \square$$

$$(d-2)(4d^2 - 7d - 2) = \square$$

$$(d-2)(4d+1) = \square.$$

$$4d+1 = \square.$$

Let  $4d+1 = k^2$ , some  $k \in \mathbb{N}$ .

$$d = m^2 + m.$$

$$d = \frac{k^2 - 1}{4}$$

$k$  odd, let  $k = 2m+1$ .

$$y = \frac{1 - (4m^2 + 4m + 1) \pm (2m+1)(2m+4)(2m-3)}{8}$$

$$= \frac{-4(m^2 + m) \pm 4 \binom{2m+1}{m+2} \binom{2m+1}{m-1}}{8}$$

$$= \frac{-m^2 - m + (2m+1)(m^2 + m - 2)}{8}$$

$$= \frac{2m^3 + 2m^2 - 4m - 2}{2}$$

$$= \frac{k^2 - 1 \pm (k^3 - 9k)}{8}.$$

$$\therefore y = \frac{-d \pm (d-2)\sqrt{4d+1}}{2} = \frac{-\left(\frac{k^2-1}{4}\right) \pm \frac{k^2-9}{4} \cdot k}{2}$$

$$= \frac{-k^2 + 1 \pm (k^3 - 9k)}{8}$$

For all  $m \in \mathbb{N}$ ,  $m \geq 2$

$$y = m^3 + m^2 - 2m - 1$$

$$x = m^2 + 2m^2 - m - 1.$$

and

$$(x, y) = (1, 1).$$

$$x = y + d$$

$$= \frac{-k^2 + 1 \pm (k^3 - 9k)}{8} + \frac{2k^2 - 2}{8} = \frac{k^2 - 1 \pm (k^3 - 9k)}{8}.$$

6. (IMO 2015 N1)

Find all  $M \in \mathbb{N}$  such that the sequence  $(a_n)_{n \geq 0}$  defined by

$$a_0 = M + \frac{1}{2} \text{ and } a_{k+1} = a_k \lfloor a_k \rfloor$$

contains at least one integer term.

Try:  $M=1$ , then  $(a_n) = \frac{3}{2}, \frac{3}{2}, \frac{3}{2}, \dots$  ✗

$M = \text{even}$ , done

$M=3$ , then  $(a_n) = \frac{7}{2}, \frac{21}{2}, 105, \dots$  ✓.

$M=5$ , then  $(a_n) = \frac{11}{2}, \frac{55}{2}, \frac{1485}{2}, 742 \cdot 1485$  ✓.

Let  $b_k = 2a_k$  for all  $k \geq 0$ .

$$\begin{aligned} \text{Then } b_0 &= 2M+1, \quad b_{k+1} = 2a_{k+1} \\ &= 2a_k \lfloor a_k \rfloor \\ &= b_k \left\lfloor \frac{b_k}{2} \right\rfloor. \end{aligned}$$

Now all  $b_k \in \mathbb{N}$ .

Suppose all  $a_k$  are not integers, so  $b_k$  are all odd, then

$$b_{k+1} = b_k \left( \frac{b_k - 1}{2} \right),$$

$$\begin{aligned} \mathbb{Z} \ni b_{k+1} - 3 &= \frac{b_k(b_k - 1)}{2} - 3 \quad \leftarrow \text{even.} \\ &= \frac{(b_k - 3)(b_k + 2)}{2} \quad \leftarrow \text{odd.} \end{aligned}$$

Suppose  
 $b_0 - 3 > 0$ .

Then  $v_2(b_k - 3) - 1 = v_2(b_{k+1} - 3)$ , so  $v_2(b_k - 3)$  decreases.

But  $v_2(b_k - 3)$  is finite, and is lower bounded. ✗

$$\therefore b_0 \leq 3; \quad 2a_0 \leq 3.$$

$$a_0 \leq \frac{3}{2}.$$

$$\Rightarrow M = 1$$

$\therefore$  All natural numbers  $M \geq 2$ ,  
satisfy the requirement.



## 7. (IMO 2016 N1)

For any  $k \in \mathbb{N}$ , denote the sum of digits of  $k$  in base 10 by  $S(k)$ . Find all polynomials  $P(x)$  with integral coefficients such that for any positive integer  $n \geq 2016$ ,  $P(n) > 0$  and

$$S(P(n)) = P(S(n)).$$

8. (IMO 2017 N1)

For each integer  $a_0 > 1$ , define the sequence  $(a_n)_{n \geq 0}$  by

$$a_{n+1} = \begin{cases} \sqrt{a_n} & \text{if } \sqrt{a_n} \text{ is an integer,} \\ a_n + 3 & \text{otherwise.} \end{cases}$$

~~3, 6, 9, 3, 6, 9, ...~~  
 3, 6, 9, 3, 6, 9, ...

We are done if  
 $a_n$  is eventually  
 periodic.

Find all values of  $a_0$  such that  $\exists A$  with  $a_n = A$  for infinitely many values of  $n$ .

① If  $a_n \equiv 2(3)$ , then for all  $m \geq n$ ,  $a_m \equiv 2(3)$ ,  
 not a perfect square, so it fails.

② Claim: if  $a_n \not\equiv 2(3)$ , and  $a_n > 9$ , then  $\exists m > n$  s.t.  
 $a_m < a_n$ .

Proof: Let  $t^2$ , the <sup>be</sup> largest square s.t.  $t^2 < a_n$ .

That means  $t \geq 3$ . The first square of

$(a_n + 3k)_{k \geq 0}$  is either  $(t+1)^2$ ,  $(t+2)^2$  or  $(t+3)^2$ .

So  $\exists m > n$  s.t.  $a_m \leq t+3 < t^2 < a_n$ . ✓

③ If  $a_n \equiv 1(3)$ , then  $\exists m > n$  s.t.  $a_m \equiv -1(3)$ .

If  $a_n = 4$ , then the sequence is 4, 2.

Proof: If  $a_n \equiv 7$ , then the sequence is 7, 10, 13, 16, 4, 2.

If  $a_n \geq 10$ , then take  $j > n$  s.t.  $a_j = \min\{a_{n+1}, a_{n+2}, \dots\}$ .

If  $a_j \equiv 1(3)$ , then  $a_j \leq 9$ . Indeed, if  $a_j > 9$ , then  
 from the claim above, there is something even smaller.

Then  $a_j = 4$  or 7. Then from results above, we are done. □.

④ If  $a_n \equiv 0(3)$ , then the whole sequence is  $\equiv 0(3)$ .

Conclusion:  
 Only multiples  
 of 3 work

If  $a_n \leq 9$ , then  $a_n = 3, 6$  or 9, and it follows a loop.  
 $\{3, 6, 9, 3, 6, 9, \dots\}$ . ✓

Otherwise, take  $j > n$  s.t.  $a_j$   
 using a similar argument,  
 $= \min\{a_{n+1}, a_{n+2}, \dots\}$ . Then  $a_j \leq 9$ . It will fall into the

loop again. ✓

9. (IMO 2018 N1)

Determine all pairs  $(a, b)$  of distinct positive integers such that  $\exists k \in \mathbb{N}$  for which the number of divisors of  $ka$  and that of  $kb$  are equal.

10. (IMO 2019 N2)

Find all  $a, b, c \in \mathbb{N}$  such that

$$a^3 + b^3 + c^3 = (abc)^2.$$

WLOG,  $a \geq b \geq c$ . Then  $a^2 \mid a^3 + b^3 + c^3$   
 $a^2 \mid b^3 + c^3$ ,  $a^2 \leq b^3 + c^3$ .

$$3a^3 \geq a^3 + b^3 + c^3 = (abc)^2$$

$$3a \geq b^2 c^2.$$

$$a \geq \frac{b^2 c^2}{3}$$

Combining,  $b^3 + c^3 \geq \frac{b^4 c^4}{9}$

$$2b^3 \geq b^3 + c^3 \geq \frac{b^4 c^4}{9}$$

$$2 \geq \frac{bc^4}{9}$$

$$c^4 b \leq 18.$$

$$c = 1,$$

then the equation becomes

$$a^3 + b^3 + 1 = a^2 b^2.$$

$$2a^3 + 1 \geq a^3 + b^3 + 1$$

$$2a^3 + 1 \geq a^2 b^2.$$

If  $2a^3 + 1 = a^2 b^2$ , then

$$a^2(b^2 - 2a) = 1.$$

$$a = 1, b^2 = 3. \quad \#$$

$$\therefore 2a^3 \geq a^2 b^2$$

$$a \geq \frac{1}{2}b^2.$$

On the other hand,  $a^3 < a^2 b^2$ ,  $a^{11} < b^2$ .

$$\therefore \frac{1}{2}b^2 \leq a < b^2.$$

If  $b \geq 5$ , then

$$a^2 \geq \frac{1}{4}b^4$$

$$\geq b^3 + 31.$$

$$\therefore (b^2 - a)a^2 \geq b^3 + 31.$$

$$b^2 a^2 - a^3 \geq b^3 + 31$$

$$b^2 a^2 \geq a^3 + b^3 + 31. \quad \#$$

$$\therefore b \leq 4.$$

If  $b = 4$ ,  $a^3 + 65 = 16a^2 \quad \phi$

If  $b = 3$ ,  $a^3 + 28 = 9a^2 \quad \phi$

If  $b = 2$ ,  $a^3 + 9 = 4a^2$

$$(a-3)(a^2 - a - 3) = 0$$

$$a = 3.$$

$\#$  If  $b = 1$ ,  $a^3 + 2 = a^2 \quad \phi$ .

$$\subseteq (a, b, c) = (3, 2, 1) \text{ and}$$

its permutations